

HEIDELBERG UNIVERSITY

IFDIFF PROJECT

Additional Theoretical Explanations for IFDIFF Examples

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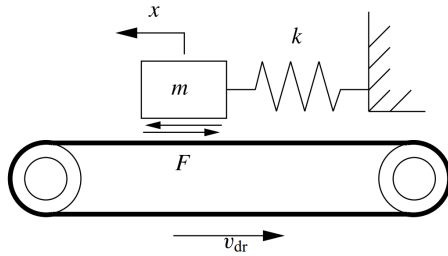
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1 A spring-loaded Stick-Slip Model

1.1 Example Overview

In the IFDIFF `friction_model` example a physical system is described, where a mass sits on a constantly moving conveyor belt and is suspended by a spring. When the spring is stretched far, its pulling force is big enough to pull back the mass and let it 'slip' over the belt. Afterwards, the mass comes to rest and 'sticks' to the belt. This creates a stable oscillatory movement. The example in its base form is formulated in [Lei+98] and a sketch, as well as an explanation of the corresponding parameters can be seen in Figure 1.



(a) System sketch taken from [Lei+98]

Quantity	Symbol	Value
Mass	m	1 kg
Spring Constant	k	1 N/m
Belt Velocity	v_b	0.2 m/s
Max Static Friction Force	F_s	1 N
Phys. Constant	δ	3.0 s/m

(b) Parameters as in [Mey20]

Figure 1: Stick-slip oscillator overview

[Mey20] gives us two switched ODE formulations (one using filippov behavior and one that does not) of second order (order of derivatives present). In each case the state of the system is described by the vector $\mathbf{x}(t) = (x_1(t), x_2(t))^T$, where x_1 is the horizontal position of the weight and x_2 is the horizontal velocity. From here we get:

$$\dot{\mathbf{x}}(t) = \begin{bmatrix} x_2(t) \\ -\frac{k}{m}x_1(t) + \frac{F(x_1(t), v_{\text{rel}}(t))}{m} \end{bmatrix} \quad (1)$$

where $v_{\text{rel}}(t) := x_2(t) - v_b$ is the mass' velocity relative to the belt. However, both formulations differ in the way they implement the function F .

The non-Filippov version uses

$$F_{\text{nF}}(x_1, v_{\text{rel}}) := \begin{cases} \min\{|kx_1|, F_s\} \text{sgn}(kx_1) & \text{if } v_{\text{rel}} = 0 \quad \text{Stick} \\ -\frac{F_s \text{sgn}(v_{\text{rel}})}{1 + \delta|v_{\text{rel}}|} & \text{if } v_{\text{rel}} \neq 0 \quad \text{Slip} \end{cases} \quad (2)$$

whereas the Filippov version uses

$$F_{\text{F}}(x_1, v_{\text{rel}}) := \begin{cases} \frac{-F_s}{1 + \delta v_{\text{rel}}} & \text{if } v_{\text{rel}} > 0 \quad \text{Slip} \\ \frac{F_s}{1 - \delta v_{\text{rel}}} & \text{if } v_{\text{rel}} < 0 \quad \text{Slip} \end{cases} \quad (3)$$

The Stick mode in the latter version is expressed as the overlap of both cases (Filippov behaviour).

1.2 A Connection between the Formulations

It can easily be checked that Equation 2 and Equation 3 coincide if $v_{\text{rel}} \neq 0$. The only thing worth explaining is the case $v_{\text{rel}} = 0$ where the mass sticks to the belt. We start by

verifying the correctness of the non-Filippov formulation.

From here on let $v_{\text{rel}} = 0$. During the "Stick" phase the mass doesn't experience any acceleration due to the constant motion of the conveyor belt. Hence, we get that $\dot{x}_2 \stackrel{!}{=} 0$ during that period of time. From here it follows that

$$0 \stackrel{!}{=} \dot{x}_2 = -\frac{k}{m}x_1 + \frac{F_{\text{nF}}(x_1, v_{\text{rel}})}{m} \quad \Leftrightarrow \quad F_{\text{nF}}(x_1, v_{\text{rel}}) = kx_1$$

and as long as $|kx_1| \leq F_s$ we get

$$kx_1 = \min\{|kx_1|, F_s\} \text{sgn}(kx_1) = F_{\text{nF}}(x_1, v_{\text{rel}})$$

(the first case in [Equation 2](#)).

In the Filippov case however, the convex combination parameter $\alpha(t) \in [0, 1]$ is chosen such that $\alpha\dot{\mathbf{x}}_- + (1 - \alpha)\dot{\mathbf{x}}_+ \parallel v_{\text{rel}} = 0$ (The notation here is as in [\[Mey20\]](#)). For the above Stick-Slip-example we can actually write α as an expression of x_1 .

Lemma 1.1. *Let $\dot{\mathbf{x}}$ be the Filippov variant of the above system. Let the rest of the notation be as above with $v_{\text{rel}} = 0$ ("Stick"-mode). Then*

$$\alpha\dot{\mathbf{x}}_- + (1 - \alpha)\dot{\mathbf{x}}_+ \parallel v_{\text{rel}} = 0 \quad \text{when} \quad \alpha(t) = \frac{F_s - kx_1(t)}{2F_s}$$

Proof. First we note that $\dot{\mathbf{x}}_-$ and $\dot{\mathbf{x}}_+$ differ only in their switch in F_{F} . Additionally, the condition $0 \stackrel{!}{=} v_{\text{rel}} = x_2 - v_b$ only places restrictions on x_2 . From this we know that $v_{\text{rel}} = 0 \perp (1, 0)^T$. The direction of Filippov sliding must therefore be $a\mathbf{e}_1$ for some $a \neq 0$.

To ensure not leaving the switching manifold, α must enforce $\alpha\dot{x}_2^- + (1 - \alpha)\dot{x}_2^+ = 0$. Taking all of this into account, the Filippov-condition reduces to:

$$\begin{aligned} 0 &\stackrel{!}{=} \alpha\dot{x}_2^- + (1 - \alpha)\dot{x}_2^+ \\ &= \alpha \left(-\frac{k}{m}x_1 + \frac{F_{\text{F}}^-(x_1, v_{\text{rel}})}{m} \right) + (1 - \alpha) \left(-\frac{k}{m}x_1 + \frac{F_{\text{F}}^+(x_1, v_{\text{rel}})}{m} \right) \\ &= -\frac{k}{m}x_1 + \frac{\alpha F_{\text{F}}^-(x_1, v_{\text{rel}}) + (1 - \alpha)F_{\text{F}}^+(x_1, v_{\text{rel}})}{m} \end{aligned}$$

From this we get:

$$\alpha F_{\text{F}}^-(x_1, v_{\text{rel}}) + (1 - \alpha)F_{\text{F}}^+(x_1, v_{\text{rel}}) \stackrel{!}{=} kx_1 \quad (4)$$

Now for $v_{\text{rel}} = 0$:

$$F_{\text{F}}^-(x_1, v_{\text{rel}}) = \frac{-F_s}{1 + \delta v_{\text{rel}}} = -F_s \quad \text{and} \quad F_{\text{F}}^+(x_1, v_{\text{rel}}) = \frac{F_s}{1 - \delta v_{\text{rel}}} = F_s \quad (5)$$

Combining [Equation 4](#) and [Equation 5](#), we can finally solve for α :

$$\alpha - F_s + (1 - \alpha)F_s \stackrel{!}{=} kx_1 \quad \Leftrightarrow \quad \alpha = \frac{F_s - kx_1}{2F_s}$$

□

Remark 1.2. In [Equation 4](#) we can see: The general Filippov condition ensures in a somewhat "natural" way, that α must be chosen such that the convex combination of F_F^- and F_F^+ matches the explicit formulation in [Equation 2](#) for the "Stick"-mode.

Remark 1.3. Again, it needs to be stressed that the expression for α as in [Lemma 1.1](#) is only analytical in terms of the numerical solution to $x_1(t)$. As of now, we do not have an analytical solution for this expression!

References

- [Lei+98] RI Leine et al. “Stick-slip vibrations induced by alternate friction models”. In: *Nonlinear dynamics* 16.1 (1998), pp. 41–54.
- [Mey20] Andreas Meyer. “Numerical solution of optimal control problems with explicit and implicit switches”. PhD thesis. 2020.